1 Between-epoch carry: arrowhead admissibility and the DD cycle basis

The Network-mode document establishes the arrowhead normal-equation topology and its exact Schur reduction (the per-station Schur reduction section of the Network-mode document), and the worked estimability example establishes the minimum-constraint datum (the two-receiver estimability example). This document develops the part that sits between them and is the hardest to get right in practice: *how information reaches an arrowhead system from outside the current epoch.*

The question is forced by the structure. A network producer that solves each epoch independently throws away everything it learned about the carrier-phase ambiguities, which are the only genuinely constant quantities in the problem; a producer that carries them naively destroys either the sparsity that makes the solve affordable or the datum invariance that makes the product meaningful. The resolution has three parts, developed in order below:

1. An arrowhead system admits exactly *three* kinds of information fold, and which one applies is decided by *which columns* the new information touches, not by what it means physically (Section 1.1).
2. The quantity that may be carried is therefore constrained to be gauge-free, and the natural gauge-free basis of the ambiguity block is the *cycle space* of the receiver–satellite observation graph — the double differences indexed by the spanning tree's *chords* (Section 1.2).
3. Carrying that basis costs one bounded Woodbury correction per connected component, with closed forms for the posterior and for an exact innovation test (Section 1.3).

Section 1.4 then carries a two-receiver, two-satellite network — the smallest network whose observation graph has a cycle at all — through every step with exact arithmetic.

Remark 1 (Theoretical footing). Time-linking a *selected* set of linear functions $\boldsymbol{z} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{x}$ while re-estimating the remainder fresh each epoch is the generalised Kalman filter of Teunissen, Khodabandeh & Psychas (*J. Geod.* 95:108, 2021) and the generalised least-squares filter of Hou & Zhang (*J. Geod.* 99:3, 2025). Both leave one question open: how to construct the transition $\boldsymbol{\Phi}^z$ of the linked functions across a *changing* linked set. Choosing $\boldsymbol{z}$ to be the cycle basis answers it, because every basis change the graph can force is then a unimodular integer matrix (Section 1.2). The spanning-tree S-basis itself is due to Jonge [1], who selects it once and never re-selects it; the construction here is what that becomes when the basis must be allowed to move.

1.1 Three doors into an arrowhead system

Write the epoch's system as in the Schur reduction section: a shared satellite border $\boldsymbol{x}_s$, one private block $\boldsymbol{x}_r$ per station, and zero station–station coupling. Suppose we wish to add a Gaussian prior $\boldsymbol{z} \sim N(\hat{\boldsymbol{z}}, \boldsymbol{P}^z)$ on some linear function $\boldsymbol{z} = \boldsymbol{C}\boldsymbol{x}$ of the epoch's parameters. In information form this is the increment $\boldsymbol{C}^{\mathrm{T}}(\boldsymbol{P}^z)^{-1}\boldsymbol{C}$ to the normal matrix and $\boldsymbol{C}^{\mathrm{T}}(\boldsymbol{P}^z)^{-1}\hat{\boldsymbol{z}}$ to the right-hand side. Where that increment *lands* decides everything.

Door 1: the shared border. If $\boldsymbol{C}$ touches only shared columns, the increment lands entirely in $\boldsymbol{N}_{ss}$. Such a fold is *reduction-preserving*:

> **Lemma.** For any symmetric $\Delta\boldsymbol{N}$ and any $\Delta\boldsymbol{u}$ on the shared border,
>
> $$\widetilde{(\boldsymbol{N}_{ss} + \Delta\boldsymbol{N})} = \tilde{\boldsymbol{N}}_{ss} + \Delta\boldsymbol{N}, \qquad \widetilde{(\boldsymbol{u}_s + \Delta\boldsymbol{u})} = \tilde{\boldsymbol{u}}_s + \Delta\boldsymbol{u}.$$
>
> *Proof.* The Schur subtrahend $\sum_r \boldsymbol{N}_{sr}^r (\boldsymbol{N}_{rr}^r)^{-1} \boldsymbol{N}_{sr}^{r\mathrm{T}}$ is a function of $\boldsymbol{N}_{sr}$ and $\boldsymbol{N}_{rr}$ only; $\boldsymbol{N}_{ss}$ enters the reduced-normals equation additively. □

The practical consequence is that a border prior may be folded *before or after* the reduction with identical results, so the between-epoch carry of the satellite clock and biases costs no second Schur pass even when it is applied late — between the null-space rank gate and the solve, say. This is what `addSharedInfo` (`filter/NetworkNormalEquations`) exploits: when a reduction is already cached it folds the delta in place and leaves the cache current. The dense form of the same carry (`addSharedPriorDense`) takes a *covariance* rather than an information matrix for a reason worth stating: a carried key absent from the current epoch's layout must be *marginalised*, which is a row/column deletion on $\boldsymbol{P}^z$, whereas deleting the same rows of $(\boldsymbol{P}^z)^{-1}$ would *condition* on the departed key and bias every survivor.

Door 2: one station's own columns, before the reduction. If $\boldsymbol{C}$ touches only station r's local columns, the increment lands in $\boldsymbol{N}_{rr}^r$. For a diagonal prior on columns $c \in \mathcal{C}$ with information weights $\Lambda_c \geq 0$ and information right-hand sides η_c,

$$\boldsymbol{N}_{rr}^r \mathrel{+}= \sum_{c\in\mathcal{C}} \Lambda_c \, \boldsymbol{e}_c \boldsymbol{e}_c^{\mathrm{T}}, \qquad \boldsymbol{u}_r^r \mathrel{+}= \sum_{c\in\mathcal{C}} \eta_c \, \boldsymbol{e}_c, \tag{1}$$

which the implementation applies as a sequence of rank-one updates of the *cached* $\mathrm{LDL}^{\mathrm{T}}$ factor (`addStationPrior`), so the raw $\boldsymbol{N}_{rr}$ never has to be stored or re-factorised and positivity of Λ_c keeps the factor positive-definite by construction. Unlike door 1 this fold is *not* reduction-preserving: it changes $\boldsymbol{N}_{rr}$, hence the Schur subtrahend, hence $\tilde{\boldsymbol{N}}_{ss}$. It must therefore be applied *before* the reduction, and there is no post-reduce fast path. Two properties make it cheap: it is orthogonal to every other station (the arrowhead has no station–station block to disturb), and it conditions only $\boldsymbol{x}_r$.

The two doors have opposite ordering constraints — door 1 may be late, door 2 must be early — and that asymmetry is not a convention but a direct reading of the reduced-normals equation.

Door 3: local columns across several stations. If $\boldsymbol{C}$ touches local columns of two or more stations, the information increment $\boldsymbol{C}^{\mathrm{T}}(\boldsymbol{P}^z)^{-1}\boldsymbol{C}$ has a nonzero (r, r') block. It is representable neither as a border increment nor as a per-station one, and folding it into the unreduced system *destroys the arrowhead*: the station–station zeros that make the marginalisation exact and cheap are exactly what it would fill in.

This is not an exotic case. It is the case that matters, because the only ambiguity combinations worth carrying are between-station ones — a single station's ambiguity is not estimable on its own (the phase part of the two-receiver example). The resolution is to fold such a prior *after* the reduction, by Woodbury through the block-diagonal local normals, using the per-station factors the reduction has already computed and left intact. Section 1.3 gives the algebra; the rest of this subsection records the two hazards it creates.

Remark 2 (Post-reduce folds do not compose). Two door-3 folds applied to the same reduced system are each derived against the *pristine* $\boldsymbol{N}_{rr}$. Composing them over coupled local columns is therefore inexact — it silently drops the cross term $\boldsymbol{C}_L \boldsymbol{N}_{rr}^{-1} \boldsymbol{C}_{L,\text{other}}^{\mathrm{T}}$ of the joint middle inverse, because neither fold sees the other's augmentation of $\boldsymbol{N}_{rr}$. The implementation makes this structural rather than advisory: the sum-to-zero ensemble phase gauge (`foldReducedSumZeroGauges`) and the carried DD priors (`foldReducedDdPriors`) are mutually exclusive, and `NetworkEstimator::solve` throws if both are set. Under the invariant carry the phase datum is realised instead by door 2 — pre-reduce tree pins — which compose with a post-reduce door 3 fold *exactly*, since the fold then reads an already-pinned factor.

Remark 3 (A door-3 fold cannot repair a rank deficiency). Let $\delta\boldsymbol{x}_s$ span a null direction of $\tilde{\boldsymbol{N}}_{ss}$, with the matching local response $\delta\boldsymbol{x}_r = -(\boldsymbol{N}_{rr}^{r})^{-1} \boldsymbol{N}_{sr}^{r\mathrm{T}} \delta\boldsymbol{x}_s$. If the carried function is estimable, $\boldsymbol{z} = \boldsymbol{C}_L \boldsymbol{x}_L$ is invariant along that direction, so $\boldsymbol{C}_L\, \delta\boldsymbol{x}_L = -\boldsymbol{G}^{\mathrm{T}} \delta\boldsymbol{x}_s = 0$ with $\boldsymbol{G}$ as defined in Equation (4). Every column of $\boldsymbol{G}$ is therefore orthogonal to the null space, the rank-K update $\boldsymbol{G}\boldsymbol{M}\boldsymbol{G}^{\mathrm{T}}$ adds nothing along it, and the declared S-basis dimension the rank gate tests is unchanged by the carry. A carry that *did* change it would be carrying a gauge, not information.

1.2 The observation graph and its cycle space

Door 3 admits only gauge-free functions, so the carried basis must be chosen accordingly. The choice is essentially forced, and it is combinatorial rather than numerical.

The graph. Fix a constellation g and a signal s. Let $G_{g,s} = (V, \mathcal{E})$ be the graph whose vertices $V = \mathcal{R} \cup \mathcal{S}_g$ are the receivers and the satellites, and whose *arcs* are the receiver–satellite pairs carrying an ambiguity this epoch: $(r, i) \in \mathcal{E}$ iff station r tracks satellite i on signal s. There is at most one arc per pair, so an unordered vertex pair is a unique arc identity (`ArcKey`, `filter/NetworkZStore`). The graph is *bipartite* by construction: no arc joins two receivers or two satellites.

Trees and chords. A connected component with n_c receivers, m_c satellites and $|\mathcal{E}_c|$ arcs has a spanning tree of $n_c + m_c - 1$ edges. The remaining

$$K_c \;=\; |\mathcal{E}_c| \;-\; (n_c + m_c - 1) \tag{2}$$

arcs are the *chords*. Adding a chord to the tree closes exactly one cycle — its *fundamental cycle*, the chord plus the unique tree path joining its endpoints — and those K_c cycles are a basis of the component's cycle space. Equation (2) is the first Betti number of the component; it is the exact number of independent double differences the component supports, and it is a property of the *graph*, not of the estimator.

The carried functions. Because the graph is bipartite, every cycle alternates receiver–satellite–receiver–satellite, so it has even length and its arcs can be signed alternately. Define, for chord c with fundamental cycle $\mathcal{C}_c$,

$$z_c \;=\; \sum_{(r,i) \in \mathcal{C}_c} \sigma_{(r,i)}\, N_{r,s}^{i}, \qquad \sigma_{(r,i)} = \pm 1 \text{ alternating}, \tag{3}$$

which for the shortest cycle (length four) is a plain double difference $N_{r_1}^{i_1} - N_{r_2}^{i_1} + N_{r_2}^{i_2} - N_{r_1}^{i_2}$, and for a longer fundamental cycle is a higher-order alternating difference on six or more columns.

> **Proposition (gauge-freeness).** z_c is invariant under both ambiguity gauges of the network model — the receiver shift $N_r^i \mapsto N_r^i - d_r$ for all i, and the satellite shift $N_r^i \mapsto N_r^i - c^i$ for all r.

Proof. A cycle visits each of its vertices exactly once, and each visited vertex is incident to exactly *two* arcs of the cycle, which by the alternating signing carry opposite signs. A shift attached to that vertex — d_r at a receiver, c^i at a satellite — therefore contributes $(+1)\delta + (-1)\delta = 0$ to Equation (3). Summing over vertices gives invariance. □

So z_c touches no shared column at all: the receiver phase bias and the satellite phase bias cancel *analytically*, not numerically. Three consequences follow immediately, and together they are the reason the cycle basis is the right carried object:

- **Estimable in every S-system.** z_c is invariant under the datum, so it may be compared across epochs whose datum differs, and graded without an S-transform (the minimum-constraint datum section).
- **Integer, and constant between slips.** z_c is an integer combination of integer ambiguities, and its process noise is $q_z = 0$ absent a cycle slip. It is the *only* part of the ambiguity block for which a zero-q time link is physically correct.
- **Basis changes are unimodular.** Two spanning trees of the same arc set give cycle bases related by an integer matrix $\boldsymbol{T}$ with $\det \boldsymbol{T} = \pm 1$, so $\boldsymbol{z}' = \boldsymbol{T}\boldsymbol{z}$, $\boldsymbol{P}^{z\prime} = \boldsymbol{T}\boldsymbol{P}^{z}\boldsymbol{T}^{\mathrm{T}}$ is an *exact* change of basis with no information loss. Basis churn — the failure mode that defeats every attempt to carry the raw ambiguities, whose individual values are datum-dependent — becomes a non-event by construction rather than by hygiene.

Graph events are exact operations. Every way the graph can change between epochs maps to an exact operation on the carried $(\hat{\boldsymbol{z}}, \boldsymbol{P}^{z})$ (`ZComponent`, `filter/NetworkZStore`):

Graph event	Operation on the store
Chord dies (arc lost)	Drop that row and column of $\boldsymbol{P}^{z}$ — pure marginalisation
Chord born (arc closes a new cycle)	Append an entry initialised by tap-off $\hat{z}_{\mathrm{new}} = \boldsymbol{f}^{\mathrm{T}}\hat{\boldsymbol{x}}$ with its conditional variance and cross-covariances
Leaf growth (arc brings in a new vertex)	Extends the tree, closes no cycle, touches no z
Tree edge dies	Pivot a covering chord into the tree ($\boldsymbol{z}' = \boldsymbol{T}\boldsymbol{z}$, $\boldsymbol{T}$ unimodular), then kill the arc as a chord. An uncovered tree edge is a cut edge: its death splits the component, which merely re-partitions existing blocks
Components merge	Union the two blocks block-diagonally (they were independent), promote one bridge arc to the tree, tap off any surplus bridges

In the producer these are not applied as a sequence of events. The per-epoch driver (`filter/NetworkInvariantCarryDriver`) is handed the current epoch's tree/chord partition from the election and computes the *composed* transform directly: a new chord's fundamental cycle lies in the previous epoch's cycle space if and only if every arc of that cycle was present last epoch, and its carried mean is then $\hat{z}_c^{\mathrm{new}} = \sum_j w_j\, \hat{z}_j^{\mathrm{old}}$ with w_j the incidence of old chord j's arc in the new cycle. Assembling $\boldsymbol{W}$ over the contributing components gives $\hat{\boldsymbol{z}}^{\mathrm{prior}} = \boldsymbol{W}\hat{\boldsymbol{z}}^{\mathrm{old}}$ and $\boldsymbol{P}^{z,\mathrm{prior}} = \boldsymbol{W}\boldsymbol{P}^{z}\boldsymbol{W}^{\mathrm{T}}$; a dead chord's column of $\boldsymbol{W}$ is zero (marginalisation), a chord whose cycle uses a newborn arc is fresh (diffuse prior, so its posterior is the data-only tap-off), and merges and splits fall out of the contributor gather with no special case.

Remark 4 (Why the tree must avoid slipped arcs). A slip on a *chord* costs exactly one carried DD. A slip on a *tree edge* corrupts every fundamental cycle routed through it. The election therefore prefers slip-free arcs for the tree (`network.phase_datum_eq34_slip_aware_election`); this is a structural argument, independent of any detector threshold.

1.3 Folding the carried double differences

Let one connected component carry K double differences with prior mean $\hat{\boldsymbol{z}}$ and covariance $\boldsymbol{P}^z$, expressed as $\boldsymbol{z} = \boldsymbol{C}_L \boldsymbol{x}_L$ where $\boldsymbol{C}_L$ is the $K \times n_L$ signed incidence matrix of Equation (3) over the epoch's local ambiguity columns, and $\boldsymbol{C}_{L,r}$ is its restriction to station r. This is a door-3 fold. Woodbury through the block-diagonal local normals gives, per component,

$$\boldsymbol{B} = \sum_r \boldsymbol{C}_{L,r} \boldsymbol{N}_{rr}^{-1} \boldsymbol{C}_{L,r}^{\mathrm{T}}, \quad \boldsymbol{G} = \sum_r \boldsymbol{N}_{sr} \boldsymbol{N}_{rr}^{-1} \boldsymbol{C}_{L,r}^{\mathrm{T}}, \quad \boldsymbol{z}_{\mathrm{free}} = \sum_r \boldsymbol{C}_{L,r} \boldsymbol{N}_{rr}^{-1} \boldsymbol{u}_r, \quad \boldsymbol{M} = (\boldsymbol{P}^z + \boldsymbol{B})^{-1}, \tag{4}$$

and the reduced system takes the rank-K correction

$$\boxed{\begin{aligned} \tilde{\boldsymbol{N}}_{ss} &\mathrel{+}= \boldsymbol{G}\,\boldsymbol{M}\,\boldsymbol{G}^{\mathrm{T}}, && (5)\\ \tilde{\boldsymbol{u}}_s &\mathrel{+}= \boldsymbol{G}\,\boldsymbol{M}\,(\boldsymbol{z}_{\mathrm{free}} - \hat{\boldsymbol{z}}). && (6) \end{aligned}}$$

Every ingredient of Equation (4) comes from the *cached* per-station $\mathrm{LDL}^{\mathrm{T}}$ factors, so no re-reduction is needed and the factors are left pristine; the only new inverse is the $K \times K$ middle matrix $\boldsymbol{M}$, which is per component and bounded. Back-substitution then returns the *constrained* locals,

$$\boldsymbol{x}_r \mathrel{-}= \boldsymbol{N}_{rr}^{-1} \boldsymbol{C}_{L,r}^{\mathrm{T}}\, \boldsymbol{M}\, \big(\boldsymbol{z}_{\mathrm{free}} - \boldsymbol{G}^{\mathrm{T}} \boldsymbol{x}_s - \hat{\boldsymbol{z}}\big), \tag{7}$$

so the ambiguities that the quality-control layer and the next epoch's tap-off read live in the same frame as the fold.

Posterior. The value the store carries forward is the shrinkage of the prior towards the data-implied DD,

$$\hat{\boldsymbol{z}}^{+} = \hat{\boldsymbol{z}} + \boldsymbol{P}^z\, \boldsymbol{M}\, \boldsymbol{v}, \qquad \boldsymbol{v} = \boldsymbol{z}_{\mathrm{free}} - \boldsymbol{G}^{\mathrm{T}} \boldsymbol{x}_s - \hat{\boldsymbol{z}}, \tag{8}$$

with covariance, in the datum-as-known shared frame $\boldsymbol{\Sigma}_{ss}$ (the solve's shared covariance with every datum-pinned and gauge column treated as exact),

$$\boldsymbol{P}^{z+} = \underbrace{(\boldsymbol{P}^z - \boldsymbol{P}^z \boldsymbol{M} \boldsymbol{P}^z)}_{\text{conditional on } \boldsymbol{x}_s} + \big(\boldsymbol{P}^z \boldsymbol{M} \boldsymbol{G}^{\mathrm{T}}\big)\, \boldsymbol{\Sigma}_{ss}\, (\boldsymbol{G} \boldsymbol{M} \boldsymbol{P}^z)\,. \tag{9}$$

The first term is the posterior given the border; the second restores the coupling through it. Although $\boldsymbol{\Sigma}_{ss}$ is datum-constrained, $\boldsymbol{P}^{z+}$ is datum-invariant, because $\boldsymbol{z}$ is estimable.

Innovation. The same factors give an exact test of the carry against the epoch's data. Under the null hypothesis that the carried DD is still valid,

$$\boldsymbol{v} = \boldsymbol{z}_{\mathrm{free}} - \boldsymbol{G}^{\mathrm{T}} \boldsymbol{x}_s - \hat{\boldsymbol{z}}, \qquad \mathrm{Cov}(\boldsymbol{v}) = \boldsymbol{M}^{-1} - \boldsymbol{G}^{\mathrm{T}}\, \boldsymbol{\Sigma}_{ss}\, \boldsymbol{G}, \tag{10}$$

and $|v_i| / \sqrt{\mathrm{Cov}(\boldsymbol{v})_{ii}}$ is the standardised statistic that flags a slipped carried DD for re-tapping. The *minus* sign in Equation (10) is not a typo and not a conservative approximation: $\boldsymbol{x}_s$ is itself prior-influenced through Equation (6), and subtracting $\boldsymbol{G}^{\mathrm{T}} \boldsymbol{\Sigma}_{ss} \boldsymbol{G}$ is precisely the correction for having tested the carry against a solution the carry helped produce. Consistency with Equations (8) and (9) is direct: $\boldsymbol{P}^z \boldsymbol{M}\, \mathrm{Cov}(\boldsymbol{v})\, \boldsymbol{M} \boldsymbol{P}^z = \boldsymbol{P}^z - \boldsymbol{P}^{z+}$.

The hard-constraint limit. Setting $K = 1$, $\hat{z} = 0$ and $\boldsymbol{P}^z \to \boldsymbol{0}$ turns Equations (5) and (6) into the rank-one fold of an exact linear constraint $\sum_r z_r = 0$ on local columns, with $\boldsymbol{M} \to \boldsymbol{B}^{-1}$. That is exactly the anchor-free sum-to-zero phase-bias gauge (`foldReducedSumZeroGauges`); the DD prior is its soft generalisation, which is the cleanest statement of why the two are the same object and cannot be composed.

Positive-definiteness, and why it must be counted. $\boldsymbol{P}^z + \boldsymbol{B}$ is positive-definite in exact arithmetic. It can fail to be so numerically — a stiff carry ($\boldsymbol{P}^z$ near zero) added to an ill-conditioned $\boldsymbol{B}$ subtracts operands of similar magnitude to reach a much smaller answer. When it fails, the only safe action is to fold *inertly*, $\boldsymbol{M} := \mathbf{0}$, which leaves the reduced system untouched and makes the posterior equal the prior.

That branch is silent by construction, and silence here is dangerous in a specific way: the component's carried sub-state stops taking data while every downstream reading of it — its formal σ, the innovation test, any ambiguity-resolution probe — continues to report on a quantity that is no longer being updated. A carry can therefore diffuse for hundreds of epochs behind clean-looking logs. The implementation consequently treats the PD screen as *always-on telemetry* rather than as an internal detail: per-fold counts of degenerate blocks and their dimension, the worst relative pivot, and the count of any repairs are exposed unconditionally (`DdFoldStats`, `filter/NetworkNormalEquations`), and an optional equilibrated factorisation with a scale-relative pivot test replaces the absolute one. The general rule this instantiates: an inert path needs a counter, because the absence of an error is not evidence that the mechanism ran.

1.4 Worked example: the smallest network with a cycle

We now carry a complete network through every step above with exact arithmetic. The network is the smallest one whose observation graph has a cycle at all: two receivers, two satellites, one signal. As in the two-receiver estimability example, everything inessential is stripped — coordinates known so the geometric range is removed, one frequency, no atmosphere, noise-free data, unit weights — so that every number below is exact. Ambiguities are in cycles.

Model and truth. With geometry and the receiver/satellite clocks removed by the code observations, the carrier phase reduces to

$$\Phi_r^i \;=\; \beta_r \;+\; b^i \;+\; N_r^i, \qquad r \in \{A, B\}, \quad i \in \{1, 2\}, \tag{11}$$

where β_r is the receiver phase bias, b^i the satellite phase bias (the producer's target), and $N_r^i \in \mathbb{Z}$ the ambiguity. Take as truth

$$\beta_A = 0.30, \quad \beta_B = -0.20, \qquad b^1 = 0.10, \quad b^2 = -0.15,$$

$$N_A^1 = 3, \quad N_A^2 = -1, \quad N_B^1 = 5, \quad N_B^2 = 2,$$

which through Equation (11) gives the four observations

Φ_r^i	$i = 1$	$i = 2$
$r = A$	3.40	−0.85
$r = B$	4.90	1.65

The graph, its tree, and its one chord. The observation graph has $n_c = 2$ receivers, $m_c = 2$ satellites and $|\mathcal{E}_c| = 4$ arcs, so by Equation (2) its spanning tree has $2 + 2 - 1 = 3$ edges and it has exactly $K_c = 1$ chord (Figure 1). Take the tree to be $\{(A, 1), (A, 2), (B, 2)\}$; the chord is then $(B, 1)$, and its fundamental cycle $B \to 1 \to A \to 2 \to B$ gives the single carried function

$$z \;=\; N_B^1 - N_A^1 + N_A^2 - N_B^2. \tag{12}$$

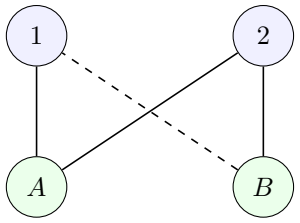


Figure 1: The smallest observation graph with a cycle: two receivers, two satellites, four arcs. Three arcs form a spanning tree; the fourth is a chord, and its fundamental cycle is the one estimable double difference Equation (12).

Two checks of the Proposition, both exact. From the truth, $z = 5 - 3 + (-1) - 2 = -1$: an integer, as it must be. From the *observations* alone, without estimating anything,

$$(\Phi_B^1 - \Phi_B^2) - (\Phi_A^1 - \Phi_A^2) = 3.25 - 4.25 = -1,$$

because β_A, β_B, b^1 and b^2 all cancel in the double difference. The single number -1 is what the carry transports; every other parameter in this section will turn out to depend on the datum.

The tree pins are the datum; the chord is left free. Model Equation (11) has $2 + 2 + 4 = 8$ unknowns and 4 observations, hence a four-dimensional null space (choose $\delta\beta_A, \delta\beta_B, \delta b^1, \delta b^2$ freely; the δN_r^i follow). Pinning the three *tree* ambiguities $N_A^1 = N_A^2 = N_B^2 = 0$ removes three of those four dimensions, leaving the single direction $(\beta_A, \beta_B, b^1, b^2) \mapsto (\beta_A + t,\ \beta_B + t,\ b^1 - t,\ b^2 - t)$ — the one remaining receiver-versus-satellite gauge, which is the network's clock gauge in miniature. It is removed by one *shared* pin, $b^1 \equiv 0$.

This is the whole datum architecture in four arcs: the tree edges are pinned through door 2 (locally, pre-reduce), the last gauge through the shared minimum-constraint datum, and the chord is left free — it is the only ambiguity that carries information rather than convention. For tidy arithmetic the pins are taken here as *exact*; the implementation realises them as tight finite priors folded by `addStationPrior`, which is why the pinned columns remain present in $\boldsymbol{N}_{rr}$ there. One consequence of the exact-pin idealisation should be named: with the tree ambiguities identically zero, $\boldsymbol{C}_L$ in Equation (12) collapses onto the chord's own column. With finite pins it retains all four entries, and the fold algebra below is unchanged in form.

Arrowhead blocks. Order the shared block $\boldsymbol{x}_s = [\,b^1, b^2\,]^{\mathrm{T}}$ and the free locals $\boldsymbol{x}_A = [\,\beta_A\,]$, $\boldsymbol{x}_B = [\,\beta_B,\ N_B^1\,]^{\mathrm{T}}$. With unit weights the blocks of the per-station block definitions are

$$\boldsymbol{N}_{ss}^A = \boldsymbol{I}_2, \quad \boldsymbol{N}_{sr}^A = \begin{bmatrix}1\\1\end{bmatrix}, \quad \boldsymbol{N}_{rr}^A = \begin{bmatrix}2\end{bmatrix}, \quad \boldsymbol{u}_s^A = \begin{bmatrix}3.40\\-0.85\end{bmatrix}, \quad \boldsymbol{u}_r^A = \begin{bmatrix}2.55\end{bmatrix},$$

$$\boldsymbol{N}_{ss}^B = \boldsymbol{I}_2, \quad \boldsymbol{N}_{sr}^B = \begin{bmatrix}1 & 1\\1 & 0\end{bmatrix}, \quad \boldsymbol{N}_{rr}^B = \begin{bmatrix}2 & 1\\1 & 1\end{bmatrix}, \quad \boldsymbol{u}_s^B = \begin{bmatrix}4.90\\1.65\end{bmatrix}, \quad \boldsymbol{u}_r^B = \begin{bmatrix}6.55\\4.90\end{bmatrix},$$

with $(\boldsymbol{N}_{rr}^B)^{-1} = \left[\begin{smallmatrix}1 & -1\\-1 & 2\end{smallmatrix}\right]$. There is no A–B block: the stations share only b^1 and b^2.

Schur reduction. Eliminating each station's locals through the reduced-normals equation gives

$$\boldsymbol{N}_{ss}^A - \boldsymbol{N}_{sr}^A(\boldsymbol{N}_{rr}^A)^{-1}\boldsymbol{N}_{sr}^{A\mathrm{T}} = \begin{bmatrix}0.5 & -0.5\\-0.5 & 0.5\end{bmatrix}, \qquad \boldsymbol{u}_s^A - \boldsymbol{N}_{sr}^A(\boldsymbol{N}_{rr}^A)^{-1}\boldsymbol{u}_r^A = \begin{bmatrix}2.125\\-2.125\end{bmatrix},$$

while station B contributes *exactly zero* to both: it has two observations and two free locals, so eliminating them absorbs its rows entirely. Summing,

$$\tilde{\boldsymbol{N}}_{ss} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}, \qquad \tilde{\boldsymbol{u}}_s = \begin{bmatrix} 2.125 \\ -2.125 \end{bmatrix}, \qquad \mathrm{eig}(\tilde{\boldsymbol{N}}_{ss}) = \{0,\, 1\}. \tag{13}$$

The reduced system has rank 1 and one zero eigenvalue, whose eigenvector is $[1, 1]^{\mathrm{T}}$ — exactly the one gauge predicted above. A null-space rank gate declaring dimension 1 for this configuration would pass; declaring 0 or 2 would throw (the minimum-constraint datum section).

Datum, solve, back-substitute. Applying the shared pin $b^1 \equiv 0$ to Equation (13) and solving gives $\hat{b}^2 = -4.25$, and back-substituting through the back-substitution equation,

$$\hat{\beta}_A = 3.40, \qquad \hat{\beta}_B = 5.90, \qquad \hat{N}_B^1 = -1.00.$$

The fit is exact (all four residuals vanish), and the result agrees to machine precision with a dense stacked solve of the same system, which is the lossless-marginalisation claim of the Schur reduction section made arithmetic.

Note how little of this resembles the truth: the estimated satellite phase bias $\hat{b}^2 = -4.25$ has nothing to do with $b^2 = -0.15$, and $\hat{N}_B^1 = -1$ has nothing to do with $N_B^1 = 5$. Each has absorbed the datum. But the chord's double difference in this frame is

$$\hat{z} = \hat{N}_B^1 - \hat{N}_A^1 + \hat{N}_A^2 - \hat{N}_B^2 = -1 - 0 + 0 - 0 = -1,$$

identical to the truth and to the value read directly off the observations. Datum-invariance is not an asymptotic property here; it is an identity.

Folding a carried DD. Now suppose the previous epoch handed us this chord. With $\boldsymbol{C}_L = [\,0 \;\; 1\,]$ on station B (one row, the exact-pin collapse noted above), Equation (4) gives

$$\boldsymbol{B} = 2, \qquad \boldsymbol{G} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad \boldsymbol{z}_{\mathrm{free}} = 3.25,$$

and the identity $z = \boldsymbol{z}_{\mathrm{free}} - \boldsymbol{G}^{\mathrm{T}}\boldsymbol{x}_s = 3.25 - 4.25 = -1$ recovers the DD, confirming that $\boldsymbol{z}_{\mathrm{free}}$ is a data-only quantity in the correction frame rather than the DD itself. Observe also that $\boldsymbol{G}^{\mathrm{T}}[1, 1]^{\mathrm{T}} = 0$: the fold direction is orthogonal to the gauge, exactly as the second Remark of Section 1.1 requires, so no choice of prior can make this carry fix the datum.

A consistent carry is a no-op. With $\hat{z} = -1$ and a rigid prior $\boldsymbol{P}^z = 0$, Equations (5) and (6) reproduce $\hat{b}^2 = -4.25$, $\hat{\beta}_A = 3.40$, $\hat{\beta}_B = 5.90$, $\hat{N}_B^1 = -1$ and zero residuals. The innovation is $v = 3.25 - 4.25 - (-1) = 0$. Carrying information the data already implies changes nothing — as it should.

A stale carry warps everything. Now suppose the chord slipped by one cycle and the carry did not notice, so $\hat{z} = 0$ is folded rigidly. Re-solving gives

	$\hat{b}^2$	$\hat{\beta}_A$	$\hat{\beta}_B$
consistent carry ($\hat{z} = -1$)	-4.25	3.40	5.90
stale carry ($\hat{z} = 0$)	-3.75	3.15	5.15

and post-fit residuals of $+0.25$, -0.25, -0.25, $+0.25$ cycles on rows $(A,1)$, $(A,2)$, $(B,1)$, $(B,2)$ respectively. A one-cycle error in a stiff carried quantity did not stay in the ambiguity that slipped: it moved the *emitted satellite phase bias* by half a cycle and smeared the remainder evenly over every observation in the network. This is the general mechanism by which a stale zero-q carry damages a solution — the network has no way to reject information asserted with infinite confidence, so it redistributes the contradiction into whatever states can absorb it. It is also why any zero-q time link requires a wired slip response before it may be armed.

The innovation test, and its limits. From Equation (10), with $\boldsymbol{\Sigma}_{ss} = \mathrm{diag}(0,\,1)$ (the pinned column treated as exact),

$$v = 3.25 - 3.75 - 0 = -0.5, \qquad \mathrm{Cov}(v) = \boldsymbol{M}^{-1} - \boldsymbol{G}^{\mathrm{T}}\boldsymbol{\Sigma}_{ss}\boldsymbol{G} = 2 - 1 = 1, \qquad \frac{|v|}{\sqrt{\mathrm{Cov}(v)}} = 0.5.$$

The test sees the slip, but at only 0.5σ — far below any sensible flag threshold. The reason is visible in the residual table: with four observations and no redundancy to speak of, half the error was absorbed by the parameters, so only half of it reached the innovation. The statistic is exact, not conservative; its *power* is a property of the network's redundancy, and a two-by-two network has almost none. This is the honest reading of any estimation-side slip test: it is diagnostic of the carry only in proportion to how much the network could not absorb.

A tree-edge death: the unimodular pivot. The two-by-two network is too small to show a pivot, because with one chord every tree edge whose death matters is a cut edge. Add a third satellite: $n_c = 2$, $m_c = 3$, $|\mathcal{E}_c| = 6$, so the tree has four edges and there are $K_c = 2$ chords. Take the tree $\{(A,1),(A,2),(A,3),(B,1)\}$ with chords $(B,2)$ and $(B,3)$, giving

$$z_1 = N_B^2 - N_A^2 + N_A^1 - N_B^1, \qquad z_2 = N_B^3 - N_A^3 + N_A^1 - N_B^1.$$

Now let the tree edge $(A,1)$ die. It is covered by both chords, so a covering chord — say $(B,2)$ — is promoted into the tree, and the dead arc is killed as a chord. Against the new tree $\{(A,2),(A,3),(B,1),(B,2)\}$ the cycle basis is

$$z_1' = N_A^1 - N_B^1 + N_B^2 - N_A^2 = z_1, \qquad z_2' = N_B^3 - N_A^3 + N_A^2 - N_B^2 = z_2 - z_1,$$

that is $\boldsymbol{z}' = \boldsymbol{T}\boldsymbol{z}$ with

$$\boldsymbol{T} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \qquad \det \boldsymbol{T} = 1.$$

$\boldsymbol{T}$ is integer with unit determinant, so $\boldsymbol{P}^{z\prime} = \boldsymbol{T}\boldsymbol{P}^{z}\boldsymbol{T}^{\mathrm{T}}$ is an exact change of basis; the carried covariance is transformed, not approximated. Killing the dead arc's row of $\boldsymbol{z}'$ is then a marginalisation, and the surviving z_2' continues with no loss. Had the carried quantity been the raw ambiguities instead of the cycle basis, this same graph event would have demanded an S-transformation of a datum-dependent state — the failure mode the cycle basis exists to eliminate.

1.5 Where each object lives

Object	Symbol	Home
Arrowhead accumulation, Schur reduction, back-substitution	`addStation`, `reduce`, `reconstructStation`	`filter/NetworkNormalEquations`
Door 1 — shared border prior	`addSharedInfo`, `addSharedPriorDense`	`filter/NetworkNormalEquations`, `filter/NetworkEstimator`
Door 2 — pre-reduce local prior	`addStationPrior`	`filter/NetworkNormalEquations`
Door 3 — post-reduce Woodbury folds	`foldReducedSumZeroGauges`, `foldReducedDdPriors`	`filter/NetworkNormalEquations`
Cycle algebra, lifecycle operations	`ArcKey`, `ZComponent`	`filter/NetworkZStore`
Per-epoch reconcile $\boldsymbol{W}$, posterior install, innovation gate	`buildDdPriors`, `installPosterior`	`filter/NetworkInvariantCarryDriver`
Posterior and innovation readouts	`ddPriorPosteriorMean`, `ddPriorPosteriorCovariance`, `ddPriorInnovation`	`filter/NetworkNormalEquations`
PD screen telemetry	`DdFoldStats`, `ddPriorBlockDegenerate`	`filter/NetworkNormalEquations`
Datum and rank gate	`NetworkSbasis`, `assertNullSpaceRank`	`filter/NetworkSbasis`

References

[1] P. J. de Jonge. “A processing strategy for the application of the GPS in networks”. PhD thesis. Delft, The Netherlands: Delft University of Technology, 1998.